🧱 Builder Basics — Block Placement Practice (Extension)

Topic: Move, Turn, Climb With Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

> 🧩 This is a bonus challenge, not a weekly lesson. Try it once you can move the agent around, before you start placing blocks by hand.

In Week 1 the agent **walked**. Here it walks and **draws** at the same time. You flip one switch — **place on move** — at the very top, and you never touch it again. After that, every step drops a block.

Your four commands:

```
move forward by 3
turn left
turn right
move up by 2
```

> 🧱 One step = one block. A turn is not a step.

1 · Predict 🔮

Read each program. The switch is ON at the top and stays ON.

```
place on move ON
set block to stone
move forward by 3
```

How many blocks land on the ground?


```
place on move ON
set block to stone
move forward by 3
turn right
move forward by 2
```

How many blocks in total? Show your adding.

Draw the shape those blocks make. Mark where the agent started with an S.

```
place on move ON
set block to stone
move forward by 2
move up by 2
move forward by 2
```

This trail does not stay flat. Describe it in one sentence.

2 · A Turn Is Not a Step 🧭

`move forward by 2` and `turn right` do very different jobs.

- `move forward by 2` — the agent **travels** two blocks. Two steps, so two blocks drop.
- `turn right` — the agent **spins** on the spot. It stays where it is. No step, so **no block**.

The switch is ON. The agent runs `turn left`. How many blocks does it place? Why?

Here are two programs. The agent starts in the same spot, facing the same way.

```
Program A            Program B
move forward by 2    move forward by 2
move forward by 2    turn right
                     move forward by 2
```

Both programs place 4 blocks. So what is different about the two trails?

A turn places no blocks — so why is `turn right` still useful when you are drawing a shape?

3 · Count the Blocks 🔢

Follow this program one line at a time. Fill in the table. Turns get a row too.

```
place on move ON
set block to stone
move forward by 4
turn right
move forward by 3
turn right
move forward by 4
```

Command	Steps	Blocks this line	Total so far
move forward by 4			
turn right			
move forward by 3			
turn right			
move forward by 4			

Two lines in this program placed zero blocks. Which ones, and why?

This trail draws three sides of a rectangle. Which side is missing, and what would finish it?

4 · Going Up 🪜

`move up by 2` is a step too — so it drops blocks as well. Up steps stack blocks on top of each other instead of laying them flat.

Mix `move forward` and `move up` and you get **stairs**:

```
place on move ON
set block to stone
move forward by 1
move up by 1
move forward by 1
move up by 1
move forward by 1
```

How many blocks does this staircase use? Show your adding.

Draw this staircase from the side. Mark the start with an S.

How would you make the staircase climb twice as high without adding more `move up` lines?

5 · Spot the Bug 🐛

Read what each program **should** do, then explain what really happens.

Bug A — This should draw a line 5 blocks long.

```
set block to stone
move forward by 5
```

What is on the ground when it finishes? What is missing from the program?

Bug B — This should draw a corner: 3 blocks, turn, 3 more blocks.

```
place on move ON
set block to stone
move forward by 3
turn right
turn right
move forward by 3
```

The agent placed 6 blocks, but there is no corner. Explain what those two turns did.

Bug C — This should be stairs going up.

```
place on move ON
set block to stone
move forward by 3
move up by 3
```

Why is this a tall pole next to a flat line instead of stairs? What would you change?

6 · Design Your Own Path 🏗️

Plan a trail before you build it. Use `move forward`, `turn left`, `turn right` and `move up`. The switch goes ON once at the top and stays ON.

Choose your block and your path. Fill in the blanks:

> My block is …
>
> move forward by … (… blocks)
>
> … (… blocks)
>
> … (… blocks)
>
> … (… blocks)

How many blocks does your whole path place? Remember the turns. Show your adding.

Draw the shape your path will make. Mark the start with an S and the end with an E.

The big idea: the switch never moved, but your trail still has corners and steps up. Explain what decides the shape of a trail, if it is not the switch.

7 · Show Your Work 📸 🎥

Open your world and build the path you designed in Section 6. Flip **place on move** ON once at the top, then never touch it again. If the shape comes out wrong, change your numbers and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.